

**Experiment 8: Conduct Z test for the Data Given using Python Objective: To test whether the average weight of a species of birds differs from 150 grams.**

```
import numpy as np

import scipy.stats as stats

sample_data = np.array([
    152, 148, 151, 149, 147, 153, 150, 148, 152, 149,
    151, 150, 149, 152, 151, 148, 150, 152, 149, 150,
    148, 153, 151, 150, 149, 152, 148, 151, 150, 153
])

population_mean = 150

sample_mean = np.mean(sample_data)

sample_std = np.std(sample_data, ddof=1)

n = len(sample_data)

z_statistic = (
    (sample_mean - population_mean)
    / (sample_std / np.sqrt(n))
)

p_value = 2 * (
    1 - stats.norm.cdf(np.abs(z_statistic))
)

print(f"Sample Mean: {sample_mean:.2f}")
print(f"Z-Statistic: {z_statistic:.4f}")
print(f"P-Value: {p_value:.4f}")

alpha = 0.05

if p_value < alpha:
    print(
        "Reject the null hypothesis: "
```

```
        "The average weight is significantly different from 150 grams."
    )
else:
    print(
        "Fail to reject the null hypothesis: "
        "There is no significant difference in average weight from 150 grams."
    )
```

**Output:**

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```
Sample Mean: 150.20
Z-Statistic: 0.6406
P-Value: 0.5218
Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.
```